

Jimin Gu, Ph. D.

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Mechanical Engineer with Ph.D. with hands-on experience designing, building, integrating, and debugging electromechanical systems. Experienced in rapid hardware prototyping, CAD-based mechanical design, motion control, camera and sensor integration, custom test fixtures, and Python-based automation. Built automated experimental platforms spanning 3D printing, wearable sensing, MEMS devices, and custom instrumentation, with strong experience in failure analysis, iterative hardware development, and cross-disciplinary system integration. 27 peer-reviewed publications.

PROFESSIONAL SKILLS

Mechanical Design & Prototyping : SolidWorks, mechanical system design, fixture and jig design, prototype assembly, design iteration, tolerance-driven design, material/process selection, rapid prototyping, vendor fabrication coordination

Mechatronics Motion control, stepper motors, Arduino/GRBL, Teensy, serial communication, camera integration, multi-light illumination, sensors, DAQ systems, BLE, electromechanical system integration

Hardware Test, Validation & Debugging: Design of experiments (DOE), prototype testing, root cause analysis, failure analysis, calibration, experimental validation, reliability testing, test fixture development

Modeling & Simulation: COMSOL Multiphysics, ANSYS FEA, FDTD, MATLAB

Data & Automation: Python (NumPy, SciPy, PyQt5), MATLAB, machine learning, DAQ systems, LabVIEW, signal processing

Engineering & Collaboration: Equipment troubleshooting, prototype development, cross-functional product integration, cross-cultural teams (Korea, US, Switzerland), industry collaboration (Samsung)

PROFESSIONAL EXPERIENCE

Post Doctoral Researcher (2023 - Current): University of California, Berkeley Professor Gu's research group

1. Binder Jet Additive Manufacturing — Hardware Integration & Automation

- Integrated and extended a custom automated binder-jet platform combining motion control, inkjet actuation, cameras, multi-source illumination, and host-side Python control.
- Developed Python-based hardware interfaces connecting the host PC with Arduino/GRBL motion control, Teensy-controlled inkjet hardware, camera acquisition, and synchronized lighting.
- Built an in-situ imaging subsystem using a camera and five-light illumination for layer-synchronized acquisition during automated printing.
- Diagnosed intermittent acquisition failures by measuring real image-write latency, identifying a timing mismatch, and redesigning the control sequence with explicit synchronization/handshaking.
- Developed experimental workflows and DOE-based process characterization linking controllable printing parameters to measurable part performance.

2. Wearable Sensor Hardware — Modeling, Prototyping & System Validation

- Designed mechanically compliant sensing structures under constraints on deformation range, manufacturability, and signal performance.
- Executed iterative development across modeling, fabrication, mechanical testing, and performance validation.
- Optimized device geometry to achieve a **0–80% working strain range** and improved sensor response.
- Integrated mechanical design constraints with sensing and electronics requirements for wearable operation.

Doctoral Student Researcher (2017 – 2023) : Korea Advanced Institute of Science and Technology (KAIST), Micro Nano Transducer Laboratory

1. MEMS-based Flexible Pressure Wearable Motion Sensor Development +3 projects

- Designed and fabricated flexible pressure-sensor hardware through iterative mechanical design, MEMS fabrication, prototype assembly, and performance validation.
- Designed a custom mechanical characterization fixture when conventional UTM tooling could not accurately test the miniature device; defined requirements, designed the fixture, coordinated CNC fabrication, assembled the system, and integrated the measurement workflow.
- Developed a multi-metric hardware validation protocol covering sensitivity, hysteresis, dynamic range, durability, resolution, noise, and environmental effects.
- Diagnosed a supplier-side machining defect during prototype scaling and implemented a design correction that preserved functional requirements and project schedule.
- Integrated mechanical sensor hardware with FPCB electronics and DAQ systems in collaboration with an industry partner.

2. Auxetic Structure-Based Strain Sensor System +2 projects

- Designed auxetic mechanical structures in SolidWorks while balancing mechanical response, manufacturability, sensing performance, and packaging constraints.
- Integrated the sensing structure with readout electronics, BLE communication, and a mobile interface across multiple prototype generations.
- Resolved mechanical fit, alignment, packaging, and sensor-interface issues through iterative prototype development.
- Developed a real-time motion-tracking demonstration integrating the wearable sensor with a Unity-based visualization environment.

EDUCATION BACKGROUND

Ph.D. in Mechanical Engineering 2019 – 2023,
Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea

M.S. in Mechanical Engineering 2017 - 2019
Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea

B.S. in Mechanical Engineering 2012 – 2017
Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea

PUBLICATIONS

(SELECTED WORKS, 27 PEER-REVIEWED ARTICLES IN TOTAL, †EQUALLY CONTRIBUTED)

1. **J. Gu**, Y. Jung, J. Ahn, J. Choi, B. Kang, Y. Jeong, J.-H. Ha, T. Kim, Y. Jung, J. Park, S. Ryu, I. Lee, I. Park, "Auxetic Kirigami Structure-Based Self-Powered Strain Sensor with Customizable Performance using Machine Learning," *Nano Energy*, vol. 130, 110124, 2024. (IF: 17.24)
2. Y. Jung†, **J. Gu†**, J. Yeo, W. Lee, H. Han, J. Choi, J.-H. Ha, J. Ahn, H. Cho, S. Ryu, I. Park, "Highly Sensitive Soft Pressure Sensors for Wearable Applications based on Composite Films with Curved 3D Carbon Nanotube Structures," *Small*, 2303981, September 2023. (IF: 13.08)
3. B. Zheng, Z. Jin, G. Hu, **J. Gu**, S. Yu, J. Lee, G. Gu, "Machine Learning and Experiments: A Synergy for the Development of Functional Materials," *MRS Bulletin*, vol. 48, February 2023. (IF: 4.1)
4. J. Ahn†, **J. Gu†**, Y. Jeong†, J.-H. Ha, J. Ko, B. Kang, S.H. Hwang, J. Park, S. Jeon, H. Kim, J.-H. Jeong, I. Park, "Nanotransfer-on-things: From Rigid to Stretchable Nanophotonic Devices," *ACS Nano*, vol. 17, no. 6, 5935–5942, 2023. (IF: 15.8)
5. **J. Gu†**, J. Ahn†, J. Jung, S. Cho, J. Choi, Y. Jeong, J. Park, S. Hwang, I. Cho, J. Ko, J. Ha, Z. Zhao, S. Jeon, S. Ryu, J. Jeong*, I. Park*, "Self-Powered Strain Sensor based on the Piezo-Transmittance of a Mechanical Metamaterial," *Nano Energy*, vol. 89, 106447, November 2021. (IF: 16.8)
6. Y. Jeong†, **J. Gu†**, J. Byun, J. Ahn, J. Byun, K. Kim, J. Park, J. Ko, J. Jeong, M. Amjadi, I. Park*, "Ultra-Wide Range Pressure Sensor based on Microstructured Conductive Nanocomposite for Wearable Workout Monitoring," *Advanced Healthcare Materials*, 2001461, March 2021. (Front Cover Article) (IF: 10.0)
7. **J. Gu**, D. Kwon, J. Ahn, I. Park*, "Wearable Strain Sensor Using Light Transmittance Change of Carbon Nanotube Embedded Elastomer with Microcrack," *ACS Applied Materials & Interfaces*, vol. 12, no. 9, pp. 10908–10917, March 2020. (Front Cover Article) (IF: 8.3)

PATENTS (KOREA)

1. I. Park, **J. Gu**, J. Ahn, J. Jeong, "Optical type strain sensor based on a mechanical metamaterial" (Application Number: 10-2021-0094119), 2021
2. I. Park, Y. Jeong, J. Ahn, **J. Gu**, "Squeegee structure and screen printer including same", Korea Patent (Application Number: 10-2019-0081581/Registration Number : 10-2293270-0000), 2019
3. I. Park, D. Kwon, **J. Gu**, "Composite for optical tension sensor, method for manufacturing the same, and optical tension sensor comprising the same", Korea Patent (Application Number: 10-1976024-0000/Registration Number: 10-1976024-0000), 2017
4. I. Park, J. Park, Y. Jeong, K. Kim, **J. Gu**, J. Lee, M. Jeon, H. Im "RF ablation needle sensing tissue pressure", Korea Patent (Application Number: 10-1969779-0000/ Registration Number: 10-1969779-0000), 2017